

Prepared reports. Faster review. Better operational context.

A SharePoint-first manufacturing intelligence platform for supervisor and manager decision support.

How people use the outputs

Shift Drafts automate collection, reconciliation, organization, and spreadsheet preparation. Supervisors review exceptions, add context, and finalize faster.

Daily Production Reports help managers understand production, molding, labor, quality, plan attainment, and downtime; investigate constraints; prioritize follow-up; and inform staffing, plan, maintenance, or coordination changes.

Weekly Production and Downtime Reports show trends, comparisons, and recurring problems for planning, improvement, meetings, reviews, and handoffs.

INPUT	PROCESS	OUTPUT	REVIEW + ACTION
Forms, SharePoint, files, machine events	Ingest, reconcile, link sources, validate	Shift drafts, daily and weekly reports, dashboard	People interpret context and decide

Impact with clear boundaries

Measured: 94.6% fewer report-replay processing rows and 86.6% smaller local PostgreSQL fixture footprint.

Estimated or modeled: about 90% faster retrieval, about 87% storage-cost reduction under linear scaling, 30 hours/week and about \$52K annual labor-value opportunity, and up to 227,500 widgets/about \$205K annual contribution-margin opportunity under documented assumptions.

The platform supports decisions. These figures are not blanket accuracy, booked savings, realized margin, autonomous decisions, or guaranteed outcomes.

Architecture and reliability evidence

SharePoint and Forms intake

Five Power Automate flows cover reminders, machine events, machine whiteboard submissions, molding-report archive, and labor notes. SharePoint lists and libraries retain operational inputs and outputs.

Source-linked reporting core

Python ingestion and reconciliation normalize evidence into eight PostgreSQL serving tables. Workbooks and the API/SPFx dashboard expose prepared reporting views with source lineage.

Fail-closed delivery

Validation preserves missing-versus-zero semantics. Artifact readback, bounded repair, one-writer releases, and rollback prevent unvalidated output from silently replacing the last known-good release.

Audience paths

Supervisors and managers: begin with output use, exceptions, and operating decisions. **Executives and recruiters:** review the concise impact and ownership narrative. **Engineers and successors:** follow the architecture, reliability, evidence, operating guide, and private dependency map.

Public/private boundary

Public assets use synthetic data and sanitized architecture. Tenant identifiers, employee data, production URLs, credentials, and exact operating details remain in the private handoff. Credentials are transferred only through an approved secure channel.